

Credit Risk Model: Business Summary

Probability of Default (PD) Model — Loan Portfolio

Prepared by: Christopher Jason Sagayaraj, Credit Risk Analysis **Date:** 15 September 2026 **Status:** Internal review draft
Sample: 58,645 applications

14.24%	0.9554	0.0008	\$93.5M
DEFAULT RATE	BEST MODEL ROC-AUC	PSI (TRAIN VS TEST)	PRINCIPAL IN DEFAULT

Bottom line

Four models were built and validated on the same held-out 20% sample. All four rank risk well. The recommendation is to **deploy the WOE/IV credit scorecard for applicant-facing decisions** and retain XGBoost as a challenger and portfolio-monitoring tool. Three things need to be flagged before anything goes live:

- The transparency premium is **real, not trivial** — the scorecard gives up 0.059 of ROC-AUC (Gini 0.793 vs 0.911) to XGBoost.
- **No model here is calibrated.** All four are fitted with balanced class weights, so predicted probabilities run roughly 2–3× above observed default rates. They are usable for ranking and cut-offs, not for expected-loss or IFRS 9 provisioning as they stand.
- Three scorecard variables carry near-zero weight and two of those have **counter-intuitive signs**. They should be dropped before sign-off.

1. PURPOSE

This model estimates the Probability of Default (PD) for loan applicants, using historical loan data to identify which applicant and loan characteristics are associated with higher default risk. The intended uses are (a) an application decision rule with an explainable score, (b) risk-based segmentation of the existing book, and (c) a monitoring baseline against which future applicant populations can be compared.

Four modelling approaches were compared: a points-based credit scorecard built on Weight of Evidence / Information Value (WOE/IV) methodology, and three benchmark machine-learning models — Logistic Regression, Random Forest and XGBoost.

2. THE DATA

The development sample is **58,645 historical loan applications** with 12 predictive attributes and a binary default outcome. There are **no missing values and no duplicate rows**. A further 39,098 applications form a scoring (test) set with no outcome labels, used for population-stability testing.

PORTFOLIO CHARACTERISTIC	VALUE	NOTE
Applications	58,645	Development sample
Defaults	8,350	14.24% of the book
Non-defaults	50,295	85.76%
Total principal advanced	\$540,563,602	Sum of loan amounts
Principal on defaulted loans	\$93,497,220	17.3% of principal — ahead of the 14.24% unit rate
Average loan — defaulted	\$11,197	Defaulters borrow ~26% more than non-defaulters
Average loan — performing	\$8,889	
Loan size range	\$500 – \$35,000	Mean \$9,218
Applicant income (median)	\$58,000	Range \$4,200 – \$1,900,000
Applicant age (median)	26	Range 20 – 84; a young book

Read-across: defaulted loans account for 17.3% of principal but only 14.24% of accounts. Loss severity therefore runs ahead of loss frequency — larger loans default more often. Any cut-off policy built on this model should be evaluated on dollars at risk, not just on the count of applications declined.

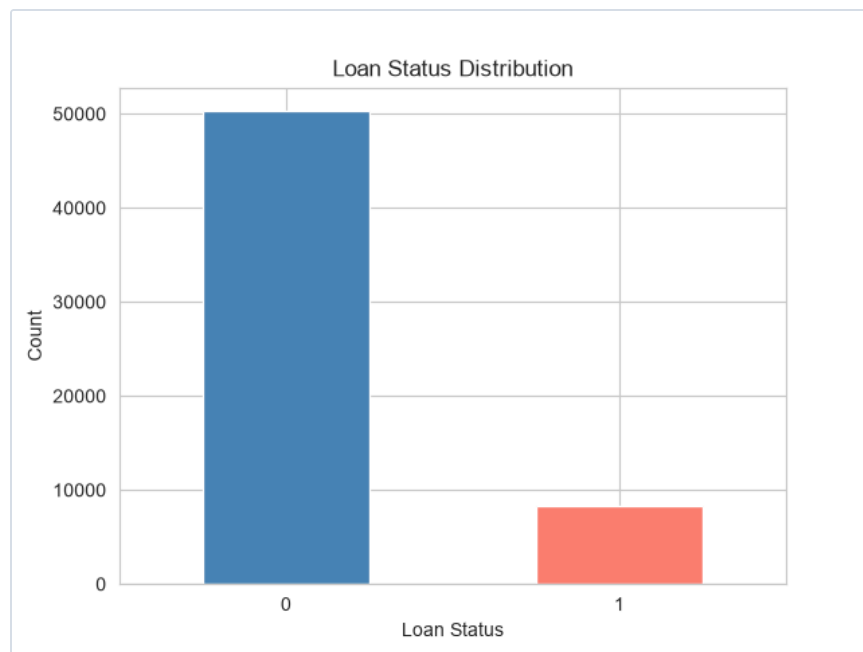


Figure 1 — Loan status distribution. 50,295 performing (0) against 8,350 defaulted (1). The roughly 86/14 imbalance is the single most important fact for how these models are trained and, as Section 4 shows, for how their accuracy figures should be read.

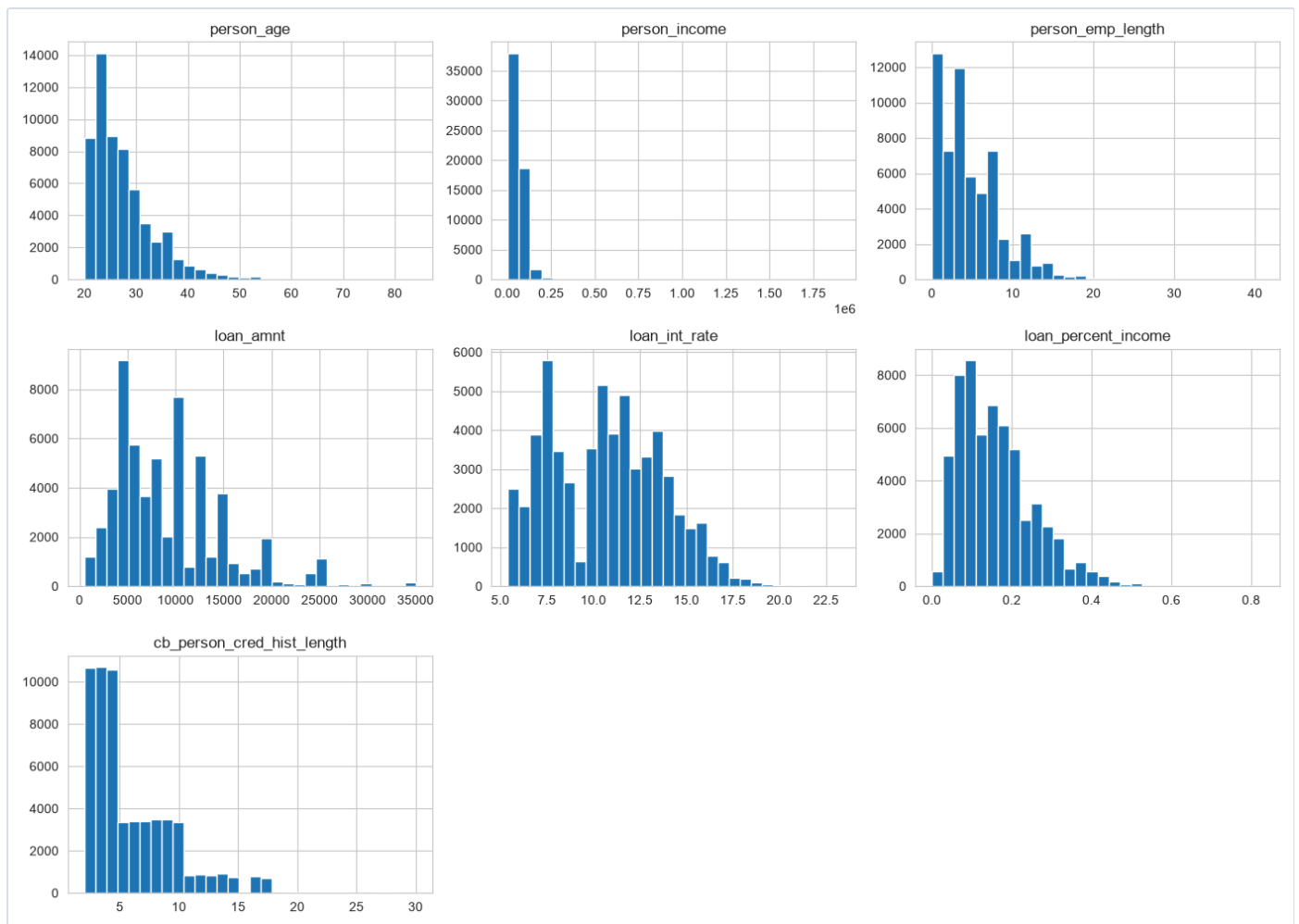


Figure 2 — Distribution of the numeric attributes. Income is severely right-skewed (a handful of applicants above \$1M compress the visible mass), age clusters in the early-to-mid twenties, and loan amount shows sharp spikes at round numbers — \$5,000, \$10,000, \$15,000 — consistent with applicants requesting headline figures rather than precisely costed amounts. The skew in income and loan-to-income is the reason the scorecard bins these variables into quantile groups rather than using them raw.

2.1 Data quality

Three records carried sentinel values of 123 : one in `person_age` and two in `person_emp_length` (i.e. 123 years old, 123 years of employment). These were corrected by median imputation, with the medians computed on the training data and the same values applied to the test set to avoid leakage. The scoring set contained no such values. After cleaning, no record shows employment length greater than or equal to the applicant's age.

Note: the original draft recorded this as "two erroneous data entries". The accurate count is three records across two fields. The correction is immaterial to any result but is stated here for the audit trail.

3. WHAT DRIVES DEFAULT RISK

Information Value (IV) measures how much each attribute separates defaulters from non-defaulters. The conventional reading is: below 0.02 useless, 0.02–0.1 weak, 0.1–0.3 medium, 0.3–0.5 strong, above 0.5 suspiciously strong and worth checking for leakage.

RANK	ATTRIBUTE	INFORMATION VALUE	STRENGTH
1	Loan grade	1.2409	Very strong
2	Loan interest rate	0.9272	Very strong
3	Loan as % of income	0.7705	Very strong
4	Home ownership	0.6009	Strong
5	Applicant income	0.4981	Strong
6	Prior default on file	0.2258	Medium
7	Employment length	0.1140	Medium
8	Loan intent	0.0951	Weak
9	Applicant age	0.0037	No predictive value
10	Credit history length	0.0036	No predictive value

3.1 Loan grade is the dominant signal

GRADE	LOANS	% OF BOOK	DEFAULTS	% OF ALL DEFAULTS	DEFAULT RATE
A	20,984	35.8%	1,032	12.4%	4.9%
B	20,400	34.8%	2,087	25.0%	10.2%
C	11,036	18.8%	1,494	17.9%	13.5%
D	5,034	8.6%	2,988	35.8%	59.4%
E	1,009	1.7%	631	7.6%	62.5%
F	149	0.3%	91	1.1%	61.1%
G	33	0.1%	27	0.3%	81.8%
All	58,645	100%	8,350	100%	14.24%

The headline in the original draft — Grade A at 4.9% against Grade G at 81.8% — is correct but understates the more useful finding. The risk does not climb smoothly across grades. It **steps**: A 4.9% → B 10.2% → C 13.5%, and then jumps to D 59.4%. Grades D through G behave as a single high-risk pool at roughly 60–80%, while A through C form a low-risk pool under 14%.

Concentration: grades D–G are **10.6% of the book but 44.8% of all defaults**, at a blended 60.0% default rate. Grades A–B are 70.6% of the book and 37.4% of defaults at a 7.5% rate. Practically, the C/D boundary is where the portfolio's risk actually lives, and that is where cut-off policy should be argued.

3.2 Home ownership is the second-strongest signal

HOME OWNERSHIP	LOANS	% OF BOOK	% OF ALL DEFAULTS	DEFAULT RATE
Rent	30,594	52.2%	81.5%	22.3%
Other	89	0.2%	0.2%	16.9%
Mortgage	24,824	42.3%	17.8%	6.0%
Own outright	3,138	5.4%	0.5%	1.4%

Renters are 52.2% of the book but carry **81.5% of every default in the portfolio**. Owners — 5.4% of the book — account for 0.5%. A 16× spread in default rate between renters and outright owners is doing a lot of work here, and it is almost certainly acting as a proxy for accumulated wealth and housing-cost stability rather than for anything causal about tenure. That is worth stating plainly because it has fair-lending implications (see Section 7).

3.3 Affordability: the 30% cliff

This is the finding I would most want to put in front of underwriting, and it is not in the original draft. Loan-to-income does not degrade gradually — it breaks.

LOAN AS % OF INCOME	LOANS	% OF BOOK	DEFAULT RATE
Under 10%	19,405	33.1%	6.7%
10 – 20%	23,767	40.5%	9.2%
20 – 30%	10,743	18.3%	14.7%
30 – 40%	3,844	6.6%	68.2%
40% and above	886	1.5%	76.2%

Default risk runs 6.7% → 9.2% → 14.7% and then leaps to **68.2%** the moment the loan passes 30% of income. That is a 4.6× step across a single band boundary. The same shape appears in interest rate — 14.0% default below a 14% rate, 43.8% above it — and in loan grade at the C/D boundary.

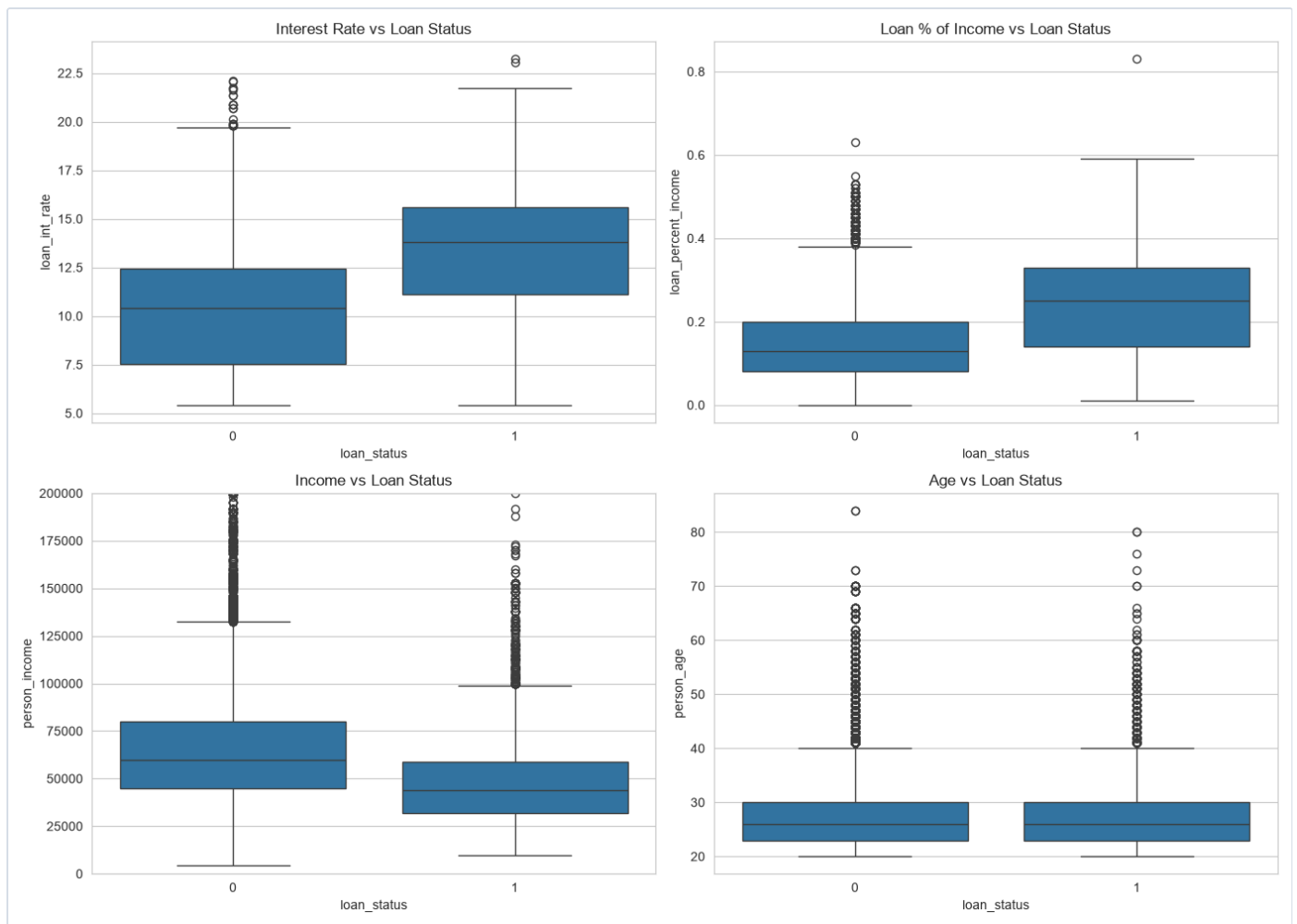


Figure 3 — Attribute distributions, defaulted (1) against performing (0). Interest rate and loan-to-income separate the two groups cleanly: defaulters sit at a median rate of ~13.9% against ~10.4%, and at ~25% of income against ~13%. Income separates in the expected direction (median ~\$44k defaulted against ~\$60k performing). Age, by contrast, shows **two essentially identical boxes** — matching its IV of 0.0037 and confirming it carries no standalone predictive signal in this population.

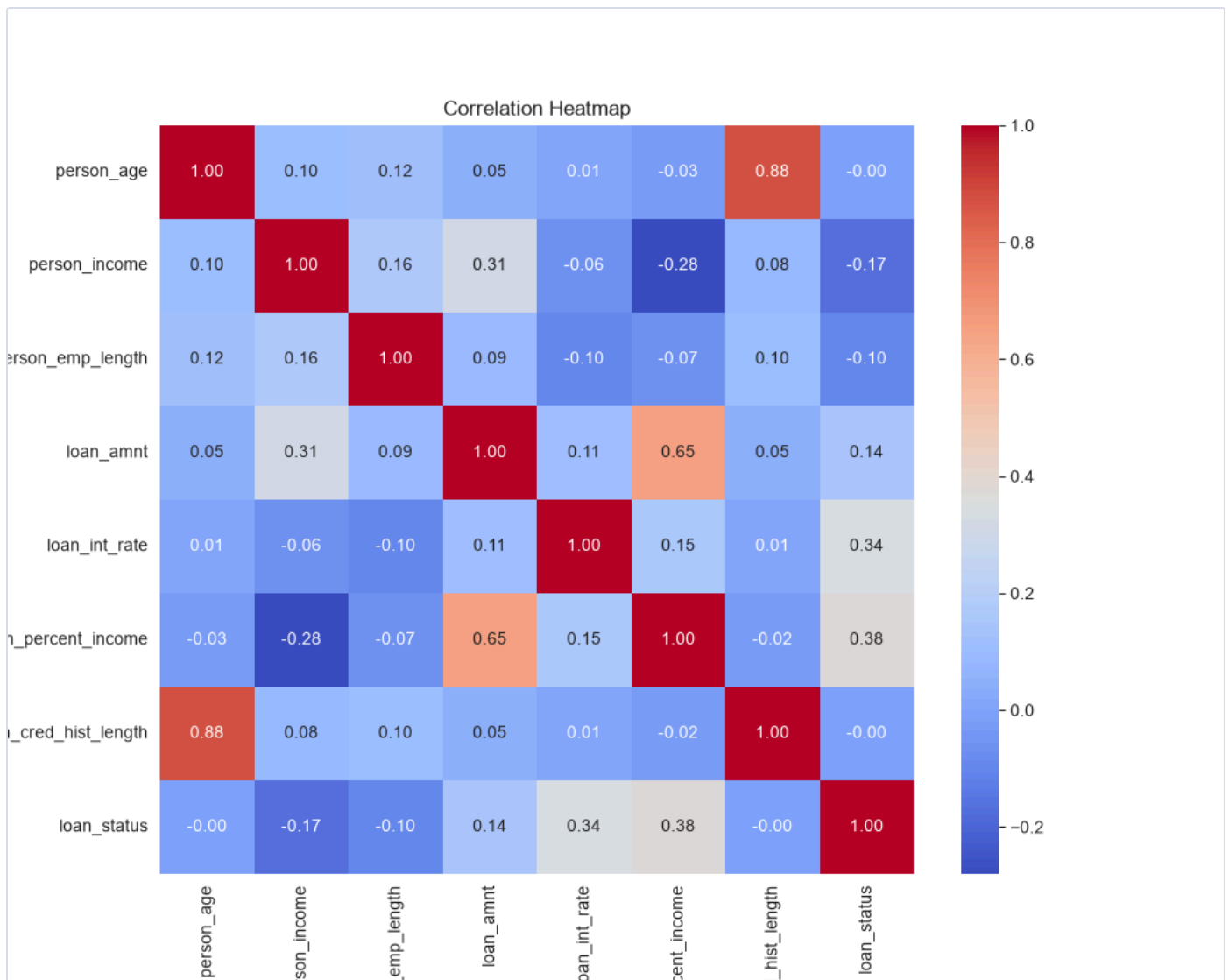


Figure 4 — Correlation between the numeric attributes and default status. Two things to note. First, the strongest linear relationships with default are loan-to-income (0.38) and interest rate (0.34) — modest coefficients that considerably understate their real power, because as Figure 3 and the 30% cliff show, the relationships are **threshold-shaped rather than linear**. This is precisely why WOE binning outperforms raw linear terms here. Second, age and credit-history length are correlated at **0.88** with each other and at 0.00 with default — they are redundant with one another *and* uninformative, and both should be dropped.

3.4 The bureau default flag adds nothing once grade is known

On its own, a prior default on file looks informative: 29.9% default rate against 11.5% for clean files, IV 0.2258. Cross-tabulated against grade, it collapses.

GRADE	NO PRIOR DEFAULT — LOANS	RATE	PRIOR DEFAULT — LOANS	RATE	DIFFERENCE
A	20,980	4.9%	4	0.0%	no population
B	20,394	10.2%	6	0.0%	no population
C	5,523	14.0%	5,513	13.1%	none
D	2,461	60.6%	2,573	58.2%	none
E	502	63.1%	507	61.9%	none
F	68	55.9%	81	65.4%	small sample
G	15	80.0%	18	83.3%	small sample

Within grades C, D and E — the only grades with meaningful numbers on both sides — applicants with a prior default actually perform *marginally better* than those without. Grades A and B contain 10 flagged applicants between them out of 41,384. The flag's apparent power is entirely an artefact of its correlation with grade: it is a grade proxy, not independent information. The scorecard's fitting process reaches the same conclusion on its own, assigning it a coefficient of 0.0304 — effectively zero.

4. MODEL PERFORMANCE

All four models were validated on the same stratified held-out 20% sample (11,729 applications, 1,670 actual defaults, 14.24% default rate — identical to the full book). The scorecard was re-fitted for this comparison with its WOE bins derived from the training portion only, so that no model sees the validation data during fitting.

MODEL	ACCURACY	PRECISION	RECALL	F1	ROC-AUC	GINI	KS	BRIER
“Approve everyone” baseline	85.76%	–	0.0%	–	0.500	0.000	0.000	–
Credit Scorecard (WOE/IV)	83.14%	44.98%	82.63%	0.583	0.8966	0.793	0.670	0.126
Logistic Regression	84.50%	47.48%	83.41%	0.605	0.9068	0.814	0.692	0.116
Random Forest	94.47%	85.33%	73.83%	0.792	0.9363	0.873	0.731	0.044
XGBoost	92.70%	70.47%	83.89%	0.766	0.9554	0.911	0.785	0.058

Accuracy is the wrong headline metric here — read AUC instead

On a 14.24% default rate, a model that simply approves every applicant scores **85.76% accuracy**. That beats both the Logistic Regression (84.50%) and the scorecard (83.14%) — despite catching precisely zero defaults and having no predictive content whatsoever. The reason those two models score below the do-nothing baseline is that both are fitted with balanced class weights, which deliberately trades false positives for catching more defaults: the scorecard declines ~26% of applicants to capture 82.6% of defaults. That is usually the right trade for a lender, but it shows up as worse "accuracy". **ROC-AUC, Gini and KS are the metrics to judge these models on**, since they measure ranking quality independent of where the approve/decline threshold is set.

4.1 Logistic Regression — accuracy 84.50%, ROC-AUC 0.9068

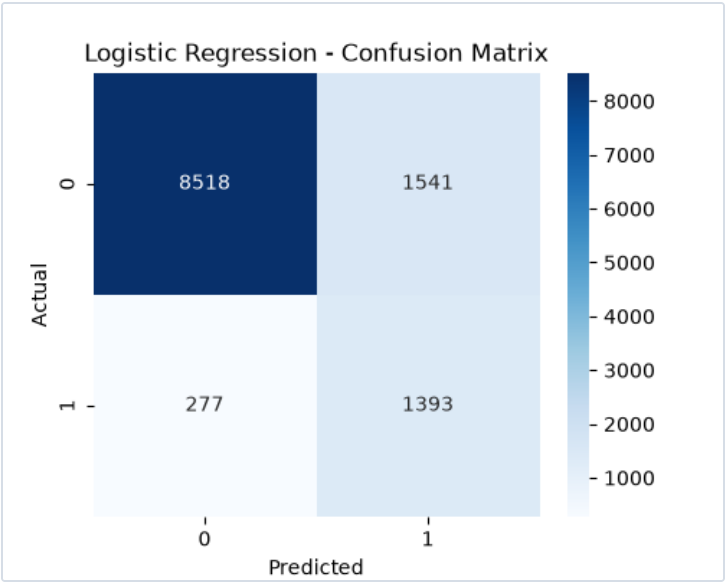


Figure 5 — Logistic Regression confusion matrix (validation set, n=11,729). Catches 1,393 of 1,670 defaults (83.4% recall) and misses 277. The cost is 1,541 false alarms — good applicants wrongly flagged — which is why precision is only 47.5%: fewer than half of everyone it flags actually defaults. As a screening filter that is defensible; as an auto-decline rule it would reject too much good business.

4.2 Random Forest — accuracy 94.47%, F1 0.792, ROC-AUC 0.9363

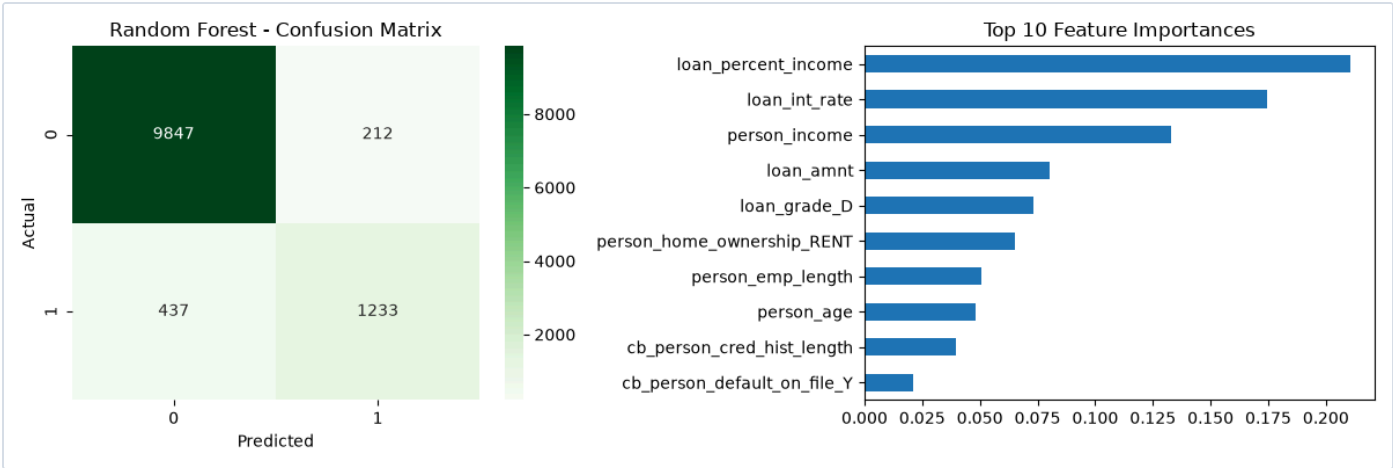


Figure 6 — Random Forest confusion matrix and top 10 feature importances. The most conservative of the four: only 212 false alarms, giving the highest precision (85.3%). But it misses **437 defaults** — 60% more than XGBoost misses — for a recall of just 73.8%. Its importance ranking is led by loan-to-income (0.209), interest rate (0.174) and income (0.133), consistent with the IV ranking in Section 3.

4.3 XGBoost — accuracy 92.70%, F1 0.766, ROC-AUC 0.9554

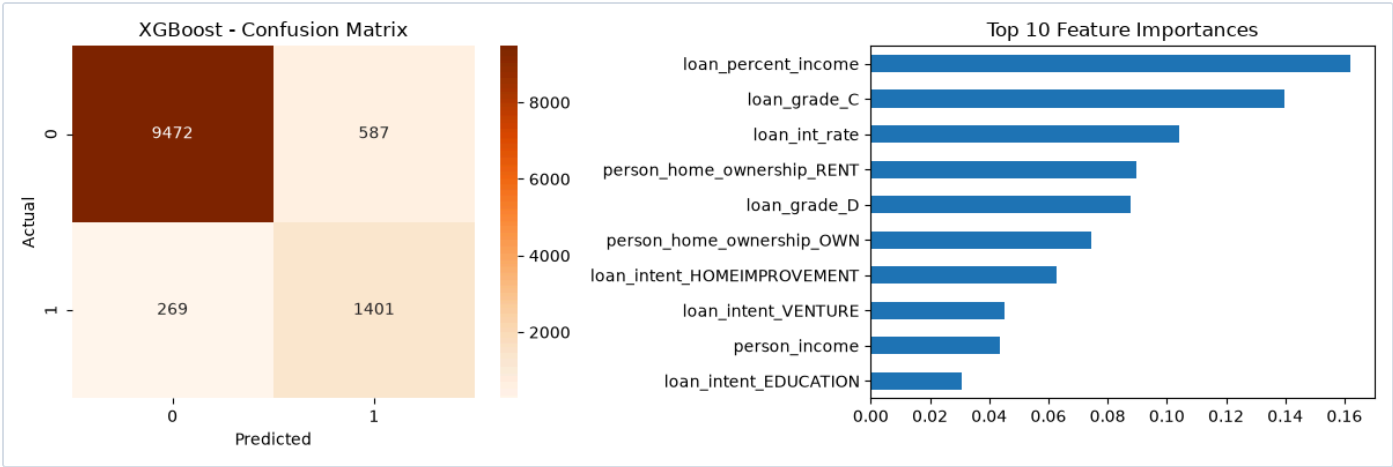


Figure 7 — XGBoost confusion matrix and top 10 feature importances. The strongest ranker of the four (AUC 0.9554, KS 0.785). It catches 1,401 of 1,670 defaults (83.9% recall) while raising 587 false alarms — a materially better trade than Logistic Regression, which achieves the same recall with 1,541 false alarms. Note that its importance ranking spreads across categorical splits (grade C, grade D, RENT, OWN, and four loan-intent categories), which makes any single feature's contribution harder to explain to an applicant than the scorecard's fixed point values.

4.4 Random Forest vs XGBoost — the choice is about which error costs more

Random Forest has the higher accuracy (94.47% vs 92.70%), but that comparison favours the wrong model for lending. Set against each other on the same 11,729 applications:

	RANDOM FOREST	XGBOOST	IMPLICATION
Defaults caught	1,233	1,401	XGBoost catches 168 more
Defaults missed	437	269	Each one is a full credit loss
Good applicants wrongly flagged	212	587	Each one is forgone margin
ROC-AUC	0.9363	0.9554	XGBoost ranks better overall

XGBoost converts 375 additional false alarms into 168 additional defaults caught. At an average defaulted loan of \$11,197 against the margin forgone on a declined good applicant, that trade is comfortably positive for almost any plausible loss-given-default assumption. **Random Forest's accuracy advantage comes from being too permissive on the class that actually costs money.**

4.5 Credit Scorecard (WOE/IV points-based model)

The scorecard scales to a base of 600 points with 20 points to double the odds — standard scaling conventions. Observed scores on the validation set run from **424 to 763**, median 640. Every point value is traceable to a single attribute band, which is what makes the model explainable to an applicant or a regulator.

ATTRIBUTE	BAND	POINTS	ATTRIBUTE	BAND	POINTS
Home ownership	Own outright	+50.9	Loan grade	A	+34.2
	Mortgage	+19.8		B	+11.0
	Other	−4.6		C	+1.7
	Rent	−11.2		D	−63.9
Loan as % of income	Up to 8%	+25.4	Income	E	−67.8
	8 – 12%	+21.1		F	−65.9
	12 – 17%	+15.5		G	−95.1
	17 – 23%	+6.2		Above \$83,119	+23.9
	Above 23%	−38.5		Below \$39,480	−18.0

Worked example — what the applicant is actually told

Base score **603** for every applicant, then add or subtract the bands that apply.

Applicant A: Grade A (+34.2), owns outright (+50.9), income \$95k (+23.9), borrowing 6% of income (+25.4), rate 7.0% (+5.6), 8 years employed (+2.6), venture purpose (+16.6), age 29 (+0.5), clean file (−0.2) → **score 762**, top band, observed default rate **1.67%**.

Applicant B: Grade D (−63.9), renting (−11.2), income \$35k (−18.0), borrowing 30% of income (−38.5), rate 15.0% (−6.0), 6 months employed (−2.4), debt consolidation (−11.7), age 22 (−0.4), prior default (+0.8) → **score 451**, bottom band, observed default rate **90.16%**. The decline letter writes itself: grade, affordability and tenure account for −114 of the −152 point gap.

Defect to fix before sign-off

Two scorecard variables carry the **wrong sign**. Prior default on file scores **+0.8 points for having defaulted before** and −0.2 for a clean file; credit history length is similarly inverted. Both are numerically trivial (under one point, against a 339-point observed score range) because the fitting correctly drove their coefficients towards zero — for the reason shown in Section 3.4 — so no live decision would turn on them. But a model reviewer or regulator will flag a scorecard that rewards a prior default, and they would be right to.

Recommendation: drop applicant age, credit history length and prior default on file from the scorecard entirely. Their combined IV is 0.23, almost all of it redundant with loan grade, and removing them eliminates the sign problem at no measurable cost to performance.

5. MODEL VALIDATION

5.1 Population Stability Index — PSI = 0.0008

PSI compares the score distribution the model was trained on against the distribution it is now being applied to. The standard reading is: below 0.10 no meaningful shift, 0.10–0.25 moderate shift worth investigating, above 0.25 significant shift and the model should be reviewed for retraining.

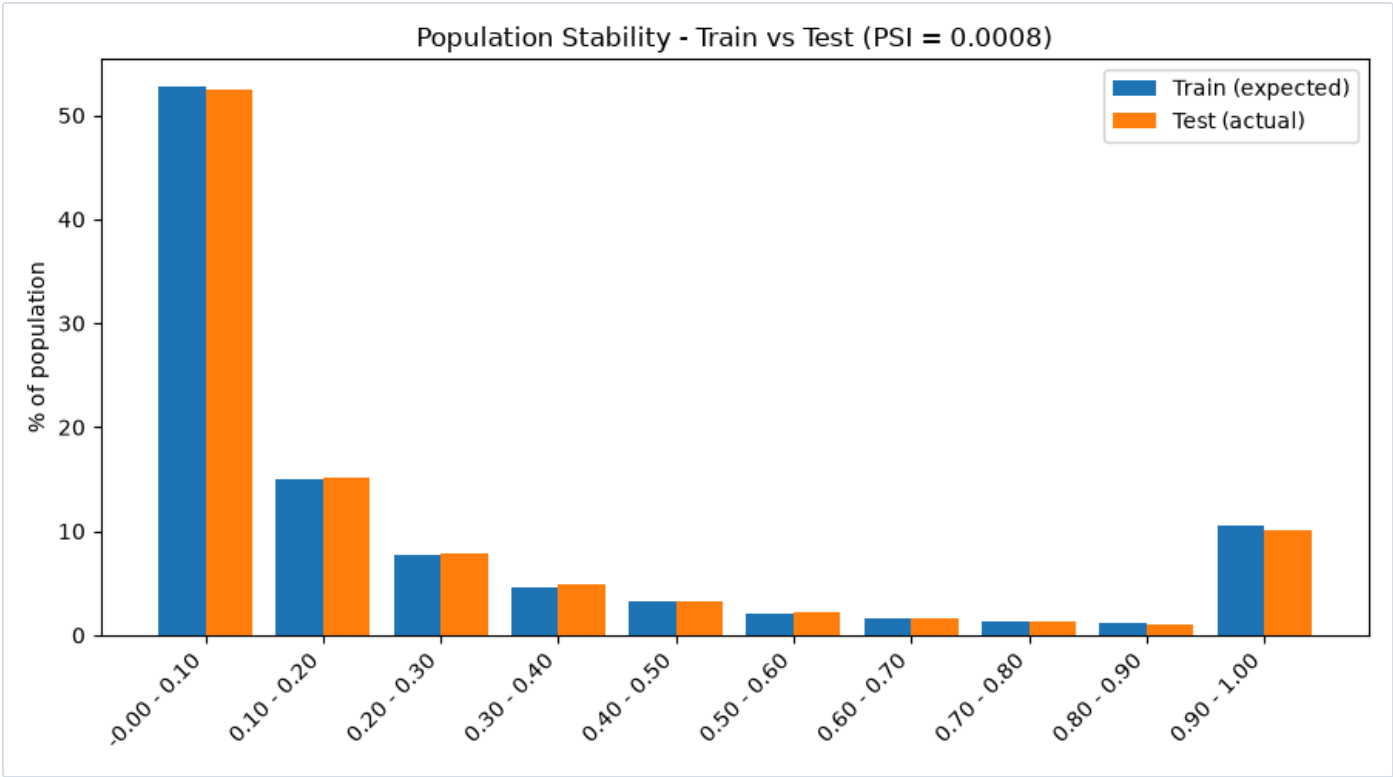


Figure 8 — Population stability, training set against scoring set. At **PSI = 0.0008**, the two populations are effectively indistinguishable — roughly 300× below the 0.25 action threshold, and the largest single-bucket contribution is 0.0003. The 39,098 applications in the scoring set are drawn from the same population the model learned on, so the model can be applied to them without adjustment.

Caveat on how much comfort to take from this. Train and test here are two random splits of one historical extract, not two periods of time. A PSI this low confirms the split was clean; it says nothing about how the applicant population will drift over the coming quarters. The real stability test is PSI measured month-on-month against live applications once the model is deployed, and that monitoring should be stood up from day one.

5.2 Calibration — the models rank well but the probabilities are wrong

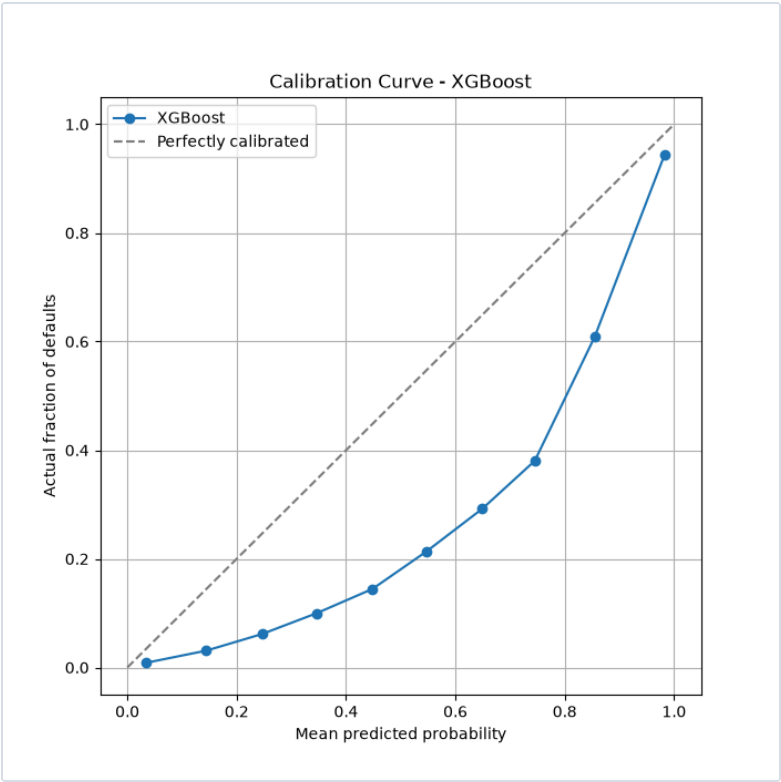


Figure 9 — Calibration curve, XGBoost on the validation set. A perfectly calibrated model tracks the dashed diagonal. This one sits **consistently below it** across the entire mid-range, meaning predicted probabilities systematically overstate the real default rate.

PREDICTED PD	ACTUAL DEFAULT RATE	GAP	READING
0.035	0.009	+0.026	~4× over
0.145	0.031	+0.114	~4.7× over
0.247	0.062	+0.185	~4× over
0.346	0.100	+0.246	~3.5× over
0.448	0.145	+0.304	~3× over
0.547	0.214	+0.333	largest absolute gap
0.649	0.292	+0.357	largest absolute gap
0.745	0.380	+0.365	largest absolute gap
0.855	0.608	+0.246	~1.4× over
0.982	0.943	+0.039	close

This is the most consequential finding in the validation section. An applicant scored at a 0.55 probability of default actually defaults 21% of the time. The cause is not a modelling error — it is the deliberate use of `scale_pos_weight` and `class_weight="balanced"`, which re-weight the minority class during fitting and inflate every predicted probability as a side effect. The same distortion applies to all four models; the Brier scores in Section 4 (0.126 for the scorecard, 0.116 for Logistic Regression) reflect it.

What this means in practice: these outputs are **ranking scores, not probabilities**. They are valid for ordering applicants, setting cut-offs and building risk bands — all of which Section 5.3 does. They are **not** valid as PD inputs to expected-loss calculations, IFRS 9 / CECL provisioning, risk-based pricing, or capital models, where an over-statement of 3× would be material. Before any such use, the model needs recalibration — isotonic regression or Platt scaling fitted on a held-out sample is the standard remedy and is a small piece of work.

5.3 Risk bands and cut-off analysis

The scorecard's practical value is that it sorts the book into bands whose observed default rates are far apart. On the validation set:

SCORE BAND	APPLICANTS	% OF PORTFOLIO	DEFAULTS	OBSERVED DEFAULT RATE
Below 500 — high risk	244	2.1%	220	90.16%
500 – 550	804	6.9%	524	65.17%
550 – 600	1,902	16.2%	630	33.12%
600 – 650	3,918	33.4%	215	5.49%
650 and above — low risk	4,861	41.4%	81	1.67%

A 54× spread between the top and bottom bands, with 75% of the portfolio sitting in the two lowest-risk bands. Translating that into a decision rule:

CUT-OFF	APPROVAL RATE	DEFAULT RATE OF APPROVED BOOK	DEFAULTS AVOIDED	GOOD LOANS DECLINED	DEFAULTING PRINCIPAL AVOIDED
No cut-off	100.0%	14.24%	0	0	\$0
500	97.9%	12.63%	220	24	\$3.2M
550	91.1%	8.67%	744	304	\$8.3M
575	84.4%	5.55%	1,121	714	\$13.0M
600	74.8%	3.37%	1,374	1,576	\$16.2M
620	63.3%	2.49%	1,485	2,820	\$17.3M
650	41.4%	1.67%	1,589	5,279	\$18.2M

Recommended operating range: a cut-off between 575 and 600. At 575 the lender declines 15.6% of applications, cuts the default rate of the approved book from 14.24% to 5.55%, avoids 67% of all defaults and \$13.0M of defaulting principal, and turns away 714 good applicants. At 600 the default rate falls further to 3.37% but the cost of good business declined more than doubles to 1,576. Above 620 the curve flattens hard — each additional default avoided costs progressively more good customers, and by 650 the lender is declining 5,279 good applicants to avoid 215 more defaults. Which point in that range is right is a commercial decision about risk appetite, not a modelling one, and it should be set with the pricing team against actual margin and loss-given-default assumptions.

6. RECOMMENDATION

Required before production

1. **Confirm the availability of loan grade and interest rate at scoring time** with underwriting. If either is assigned after the credit decision, both must be removed and every performance figure in this report re-measured. This is the single largest open risk to the results.
2. **Recalibrate.** Fit isotonic regression or Platt scaling on a held-out sample so the outputs are usable as probabilities, not just as ranks. Required before any provisioning, pricing or capital use.
3. **Drop age, credit history length and prior default on file** from the scorecard, resolving the wrong-sign defect.
4. **Run a fairness review** (Section 7) before the model touches a live application.
5. **Stand up monthly PSI and outcome monitoring** from the first day of deployment.

7. LIMITATIONS

- **Single historical snapshot.** The model is trained and validated on one extract, split randomly. There is no out-of-time validation, so nothing here demonstrates stability across economic conditions. The PSI of 0.0008 confirms the random split was clean; it does not evidence through-the-cycle robustness. An out-of-time holdout should be the first addition to this work.
- **Possible target leakage in the two strongest predictors.** Loan grade (IV 1.24) and interest rate (IV 0.93) are lender-assigned. If they encode a risk assessment made at or after underwriting, the model is partly predicting its own inputs and real-world performance will fall short of the figures reported here.
- **No fairness or disparate-impact testing has been performed.** This is the most serious outstanding gap. The dataset carries no protected characteristics, so none can be tested directly — but home ownership, the second-strongest predictor, is a well-documented proxy for wealth and is correlated with race and age in most lending populations. Renters receive -11.2 points against +50.9 for outright owners: a 62-point swing, which at 20 points to double the odds means the scorecard treats a renter as roughly 8× the odds of default of an outright owner before any other attribute is considered. That needs disparate-impact testing against the lender's own applicant demographics before deployment, and a documented business-necessity justification if the effect is confirmed.
- **Three corrected records.** One `person_age` and two `person_emp_length` values of 123 were median-imputed. Immaterial at this sample size, but the presence of sentinel values suggests the upstream data feed has no range validation, which is worth raising with data engineering.
- **Young, renter-heavy population.** Median applicant age 26, 52.2% renting. Conclusions may not transfer to an older or more asset-heavy applicant base without revalidation.